Primary-source identification under heterogeneous concept drift

Ten-site benchmark on ACS covariates

This note is the experiment section for the manuscript. It can be read on its own. The file `sec_primary_driver.tex` is written to be input after the federated-aggregation appendix. The section defines its own notation. The split used throughout is the manuscript's: covariate shift is a change in $P(X)$, and concept drift is a change in $P(Y \mid X)$ between D_{exist} and D_{new}.

1 Primary-source identification across heterogeneous sites

The manuscript separates covariate shift, a change in $P(X)$, from concept drift, a change in $P(Y \mid X)$ between a reference batch D_{exist} and a new batch D_{new}. Feature-level procedures in the paper, including the pseudo-outcome risk and conditional permutation importance, score coordinates inside one batch. The federated aggregation in Appendix 11 then asks whether those variable-importance vectors can be merged without moving raw rows. That merge treats every site as a noisy look at one common shift. A prior operational question is which sites are the primary sources of the shift, and which sites are either stable or actively destroying the consensus. This section constructs a ten-site benchmark in which the answer is known, and evaluates a counterfactual concept score against that answer.

The design isolates concept drift. Every site is a random slice of one real covariate table, so the marginal distribution $P(X)$ is shared. The American Community Survey extracts below are cross-sections, not panels, so the batch index is a seeded random order rather than a clock. Stationarity of $P(X)$ is then a construction, and a covariate arm that fires is evidence of a bug in the generator rather than a feature of the task. Labels follow the real outcome through a burn-in and are then rewritten, on a known subset of sites, by a known rule.

1.1 Data and the ten-site generator

Two WhyShift tasks supply the base table (X, y^{base}), both taken from the 2018 ACS one-year person file for California and both using the official `folktables` feature lists. ACS Income has $p = 10$ covariates and the indicator that personal income exceeds \$50,000. ACS Public Coverage has $p = 19$ covariates and the indicator of public health-insurance coverage. In each Monte Carlo replicate the analysis sample is winsorized at the first and ninety-ninth percentiles, standardized, and cut into $K = 10$ non-overlapping sites of $n_k = 1600$ rows. Each site is then cut into $T = 16$ batches of 100 rows. The burn-in is the first four batches. The concept change, when it is switched on, begins at batch $t^\star = 8$.

The ten sites have four roles. Four clean sites keep $y = y^{\text{base}}$ at every batch. Three linear sites rewrite labels inside an upper tail of a designated coordinate X_1. Two nonlinear sites rewrite every activated row. One noise site replaces activated labels with independent Bernoulli(1/2) draws. The noise site is a consensus breaker. It is not a primary source: there is no new concept for a local model to learn. Table 1 records which coordinates carry the rules. Income uses hours worked, educational attainment, and age. Public Coverage uses age, educational attainment, and personal income. These

Table 1: Rule coordinates. Z_j denotes the analysis-sample standardize of the named column. Linear sites rewrite only the upper tail of Z_1. Nonlinear sites rewrite every activated row.

Task	Role of (X_1, X_2, X_3)	X_1	X_2	X_3
ACS Income	hours, schooling, age	WKHP	SCHL	AGEP
ACS Public Coverage	age, schooling, income	AGEP	SCHL	PINCP

are the continuous or ordinal columns in each official feature list. The remaining columns stay in the table as nuisance covariates.

Write Z for the standardized covariates and let $M_k(q) = \{Z_1 > Q_q(Z_1)\}$ be the upper tail at quantile q, computed on the pooled analysis sample. The main design uses $q = 0.5$. On a linear site the candidate label is the halfspace

$$y^{\text{rule}} = \mathbf{1}\{\sigma_k(\beta_{k1}Z_1 + \beta_{k2}Z_2) > c_k\} \quad \text{on } M_k(q), \tag{1}$$

and y^{base} off that tail. The three coefficient vectors are $(1, 0.35)$, $(0.25, 1)$, and $(-0.85, 0.55)$. The threshold c_k is the quantile of the linear score that matches the positive rate of y^{base} on $M_k(q)$, and the sign $\sigma_k \in \{\pm 1\}$ is the orientation whose matched labels agree less with y^{base}. The sign choice is part of the generator. It keeps the planted concept from collapsing back onto the real outcome when hours or schooling already predict income. It is not estimated by the detector.

On a nonlinear site the same rate match is applied to every row, not only the tail:

$$y^{\text{rule}} = \mathbf{1}\{Z_1 Z_2 > c\} \qquad \text{or} \qquad y^{\text{rule}} = \mathbf{1}\{\sin(Z_1) + Z_3^2 > c\}. \tag{2}$$

The first site uses the product and the second uses the sine-square rule. The planted feature support is $\{X_1, X_2\}$ for every linear site and for the product site, and $\{X_1, X_3\}$ for the sine-square site.

Activation is a batch weight $w_t \in [0, 1]$. Under abrupt activation, $w_t = \mathbf{1}\{t \geq t^\star\}$. Under gradual activation, $w_t = 0$ before $t^\star$ and then rises linearly to one over four batches. Each post-burn-in row is rewritten with probability αw_t, and otherwise keeps y^{base}. The main grid uses $\alpha = 1$. The noise site uses the same weights and then draws a fresh coin flip. Clean sites ignore the weight.

Two consequences are worth stating before any score is computed. First, because the structured rules match the local positive rate, a marginal label discrepancy is not a substitute for concept drift. The largest average gap between the post-change positive rate and the baseline, among structured sites, is 2.4 percentage points (Income, nonlinear). Second, hours worked in the ACS piles up at 40. The empirical quantiles of WKHP at 0.5 and at 0.7 are both 40, so $\{Z_1 > Q_{0.5}\}$ and $\{Z_1 > Q_{0.7}\}$ are the same set, about 23% of Income rows. The median tail in the main Income design is therefore thinner than a continuous coordinate would suggest. Public Coverage uses age as X_1, and that tail is about half the sample. Table 2 records the realized flip rates. Structured sites move roughly half the labels inside the region they rewrite, and they leave the marginal positive rate near the baseline. The noise site flips about half of all post-burn-in labels and shifts the positive rate to one half.

1.2 The counterfactual concept score

Let $\mathcal{D}_0$ be the pooled burn-in, eight thousand rows in the main design, still generated from y^{base} at every site. Let $\mathcal{D}_k$ be the most recent six batches at site k, six hundred rows, all of them inside the drift window when activation is abrupt. A random forest with 25 trees, depth 8, and minimum leaf size 5 is the outcome model throughout, in line with the recommendation that the forest be used for distance estimation rather than for a causal reading of the meta-learner. Local predictions on $\mathcal{D}_k$ are three-fold cross-fit probabilities μ_k^{oof}. The reference probabilities μ_0^{sm} are an average of two

Table 2: Realized label changes after $t^\star$, averaged over four seeds at $\alpha = 1$, abrupt activation, and $q = 0.5$. Flip rate is the fraction of post-burn-in rows whose label differs from y^{base}. Agreement is the fraction of the rewritten region on which the rate-matched rule reproduces y^{base}. Clean sites are not rewritten, so agreement is one. The noise site has no rule, so agreement is left blank.

Task	Site type	Flip rate	Rule agreement	Post positive rate	Base positive rate
Income	Clean	0.000	1.000	0.406	0.406
Income	Linear	0.131	0.427	0.400	0.414
Income	Nonlinear	0.555	0.436	0.396	0.420
Income	Noise	0.498	—	0.491	0.410
Public Coverage	Clean	0.000	1.000	0.369	0.367
Public Coverage	Linear	0.259	0.476	0.378	0.372
Public Coverage	Nonlinear	0.492	0.506	0.371	0.374
Public Coverage	Noise	0.496	—	0.495	0.362

forests, each trained on a random subset of $\mathcal{D}_0$ whose size equals the local training set, two thirds of the current window. Matching the training size stops the larger reference sample from looking like a concept change. A third forest trained on all of $\mathcal{D}_0$ is retained only as the pooled excess defined below. It is not the headline score.

On a region $\mathcal{R}$ that has both classes in the current window and in the reference, at least fifty current rows, and at least eighty reference rows, write $B(y, p)$ for the Brier score and

$$\Delta_k(\mathcal{R}) = \Big(B(\mu_0^{\text{sm}}) - B(\mu_k^{\text{oof}})\Big)_+ \cdot \Big(2\big(\text{AUC}(\mu_k^{\text{oof}}) - \tfrac{1}{2}\big)\Big)_+. \tag{3}$$

The first factor is the excess risk of the reference concept over a local concept of the same capacity. The second factor is zero when the local model does not beat a coin, and it equals one when the local AUC equals one. The headline score is the maximum over a dictionary that does not use labels,

$$\Delta_k^{\text{cf}} = \max_{\mathcal{R} \in \{\text{all, high, low}\}} \Delta_k(\mathcal{R}), \tag{4}$$

where "high" and "low" are the two sides of the burn-in median of X_1. The median is computed on $\mathcal{D}_0$ only. The whole-window score $\Delta_k(\text{all})$ is reported beside the maximum so that the cost of ignoring the tail is visible.

The same two probability vectors define the concept contrast

$$\tau_k^2 = \frac{1}{|\mathcal{D}_k \cap \mathcal{R}_k^\star|} \sum_{i \in \mathcal{D}_k \cap \mathcal{R}_k^\star} \big(\mu_k^{\text{oof}}(X_i) - \mu_0^{\text{sm}}(X_i)\big)^2, \tag{5}$$

with $\mathcal{R}_k^\star$ the region attaining the maximum in (4). This is the concept contrast that the pseudo-outcome decomposition isolates from the covariate arm once $P(X)$ is shared across D_{exist} and D_{new}. The benchmark does not re-estimate a propensity: under the common covariate table that weight is not where the signal sits. Gating by the same skill factor as in (3) produces $\tau_k^{2,\text{g}}$. Raw τ_k^2 is retained because a flat local predictor on a noise site can still sit far from a sharp reference predictor. The gate is the part of Δ_k^{cf} that refuses to call that gap a primary source.

For a score s and two disjoint collections of sites $\mathcal{A}$ and $\mathcal{B}$, the pairwise concordance is

$$C(s; \mathcal{A}, \mathcal{B}) = \frac{1}{|\mathcal{A}|\,|\mathcal{B}|} \sum_{a \in \mathcal{A}} \sum_{b \in \mathcal{B}} \mathbf{1}\{s_a > s_b\}. \tag{6}$$

Table 3: Mean score by site type at $\alpha = 1$, abrupt activation, and $q = 0.5$. Four seeds. Δ^{cf} is (4). Δ(all) is the whole-window score. Δ(high) is the upper tail of X_1. Domain AUC is the logistic classifier on X only.

Task	Type	Δ^{cf}	Δ(all)	Δ(high)	τ^2	Skill	Domain AUC
Income	Clean	0.003	0.000	0.002	0.015	0.696	0.498
Income	Linear	0.253	0.024	0.253	0.181	0.937	0.482
Income	Nonlinear	0.336	0.298	0.316	0.313	0.897	0.494
Income	Noise	0.001	0.000	0.000	0.096	0.008	0.494
Public Coverage	Clean	0.001	0.000	0.001	0.014	0.364	0.503
Public Coverage	Linear	0.200	0.052	0.200	0.132	0.943	0.493
Public Coverage	Nonlinear	0.236	0.182	0.179	0.159	0.970	0.478
Public Coverage	Noise	0.003	0.002	0.003	0.046	0.089	0.528

Ties count as failures. The primary set $\mathcal{P}$ is the five structured sites. The target for the headline score is $C(\Delta^{\mathrm{cf}}; \mathcal{P}, \mathcal{C}) > 0.8$, where $\mathcal{C}$ is the set of four clean sites. The same functional is reported for $\mathcal{P}$ against the noise site, and separately for the linear and nonlinear subsets of $\mathcal{P}$. Every entry below is a mean over four seeds, with the worst seed in parentheses, unless a row is explicitly a per-type mean.

Negative controls use the same windows and never see the label rule. The domain AUC is the cross-fit AUC of a classifier trained to separate burn-in rows from recent rows using X alone, once with the forest and once with logistic regression. The covariate discrepancy is an RBF MMD on subsamples of size 250, with bandwidth equal to the median of the squared Euclidean distances. A joint classifier two-sample test appends y to X before the same forest AUC is computed. Marginal label total variation is the absolute difference of positive rates between the recent window and the burn-in.

Table 3 is the raw material of the concordance claim. On both tasks, and for every linear site, the maximum is exactly the upper-tail score: $\Delta^{\mathrm{cf}} = \Delta(\mathrm{high})$. On Income that tail score is 0.253 against a whole-window score of 0.024. The whole window is mostly unrewritten rows, and a single Brier average dilutes a real halfspace by about an order of magnitude. The noise skill is 0.008 on Income and 0.089 on Public Coverage, so the gate shuts even though the ungated contrast τ^2 on Income noise (0.096) exceeds the clean contrast (0.015). Logistic domain AUCs sit on 1/2 for every type. The largest is 0.528, on the Public Coverage noise site. The forest domain AUC, not shown column by column, has type means between 0.490 and 0.514, and equals 0.514 on that same noise site. Neither is evidence of a covariate shift.

1.3 Concordance

Table 4 is the primary result. On both tasks and in every seed, $C(\Delta^{\mathrm{cf}}; \mathcal{P}, \mathcal{C}) = 1$, and the same equality holds for linear sites alone, for nonlinear sites alone, and for $\mathcal{P}$ against the noise site. The whole-window score and the skill-gated contrast $\tau^{2,\mathrm{g}}$ match this ranking at full intensity. Figure 1 plots the clean-site comparison. The covariate rows do not. Random-forest domain AUC and RBF MMD have concordances between 0.38 and 0.66, with worst seeds as low as 0.17. Label total variation is consistent with a matched positive rate: its concordance against clean sites is 0.50 on Income and 0.61 on Public Coverage, and its concordance against the noise site is 0.05 and 0.00. The noise site is the unique marginal shift, and a marginal functional ranks it first.

The joint classifier two-sample test sees any change in (X, Y). It separates $\mathcal{P}$ from $\mathcal{C}$ at 0.85 on Income and 0.95 on Public Coverage, which clears 0.8 on average, but it ranks the noise site as

Table 4: Pairwise concordance (6) at $\alpha = 1$, abrupt activation, and $q = 0.5$. Each cell is the mean over four seeds, with the worst seed in parentheses. $\mathcal{P}$ is the five structured sites and $\mathcal{C}$ is the four clean sites.

Task	Score	$\mathcal{P} > \mathcal{C}$	$\mathcal{P} >$ noise	Linear $> \mathcal{C}$	Nonlinear $> \mathcal{C}$
Income	Δ^{cf}	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)
Income	Δ(all)	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)
Income	$\tau^{2,\mathrm{g}}$	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)
Income	Pareto rank	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)
Income	τ^2	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)
Income	Pooled excess	1.000 (1.000)	0.400 (0.400)	1.000 (1.000)	1.000 (1.000)
Income	Label TV	0.500 (0.100)	0.050 (0.000)	0.500 (0.083)	0.500 (0.125)
Income	Joint C2ST	0.850 (0.800)	0.350 (0.200)	0.750 (0.667)	1.000 (1.000)
Income	RF domain AUC	0.487 (0.400)	0.500 (0.000)	0.521 (0.333)	0.438 (0.250)
Income	RBF MMD	0.412 (0.300)	0.500 (0.000)	0.375 (0.167)	0.469 (0.125)
Public Coverage	Δ^{cf}	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)
Public Coverage	Δ(all)	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)
Public Coverage	$\tau^{2,\mathrm{g}}$	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)
Public Coverage	Pareto rank	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)
Public Coverage	τ^2	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)
Public Coverage	Pooled excess	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)
Public Coverage	Label TV	0.613 (0.350)	0.000 (0.000)	0.604 (0.417)	0.625 (0.250)
Public Coverage	Joint C2ST	0.950 (0.850)	0.450 (0.200)	0.917 (0.750)	1.000 (1.000)
Public Coverage	RF domain AUC	0.438 (0.350)	0.200 (0.000)	0.396 (0.167)	0.500 (0.250)
Public Coverage	RBF MMD	0.600 (0.500)	0.200 (0.000)	0.562 (0.417)	0.656 (0.375)

the stronger departure: concordance of $\mathcal{P}$ against noise is 0.35 and 0.45. On Income the linear-only comparison for the joint test falls to 0.75. A test of the joint distribution is not a test for a learnable new concept.

Pooled excess makes the same mistake in a sharper way, and only on the task where the reference concept is easy. It is the Brier gap between the forest trained on the entire burn-in and the local cross-fit model, before size matching and before the skill gate. On Income its concordance against the noise site is 0.400 in every seed. That fraction is exact: both nonlinear sites beat the noise site, and none of the three linear sites do. The pooled Income forest is a sharp predictor of y^{base}. Evaluated on coin-flip labels its Brier rises by 0.084 on average, while the same forest evaluated on a linear tail that covers 23% of the window rises by only 0.033. Size matching does not by itself fix the comparison. On the region that attains Δ^{cf}, the size-matched excess on Income noise remains 0.071, against 0.002 on clean sites. Multiplying by the skill factor is what sends the noise score from 0.071 to 0.001. Public Coverage does not reproduce the 0.40 figure. There the baseline concept is harder, the pooled excess on noise is 0.035, and every structured site still outranks it. The failure mode is real on the task where the reference fit is strong, which is the task on which an ungated risk gap is most tempting.

The Pareto comparison uses three axes when the time curve below is available, and two otherwise. The axes are Δ^{cf}, $\tau^{2,\mathrm{g}}$, and the time jump, defined as the maximum of the curve on endpoints at or after $t^\star$ minus the maximum on endpoints before $t^\star$. A site's rank is one plus the number of sites that are at least as large on every axis and strictly larger on at least one. Concordance is computed on the negative rank, so a smaller rank is a higher score. Mean ranks on Income are 1.875 for nonlinear sites, 3.250 for linear sites, 6.625 for clean sites, and 6.750 for the noise site. On Public Coverage they are 1.375, 2.667, 7.188, and 6.500. The undominated front is mostly the nonlinear sites: half of

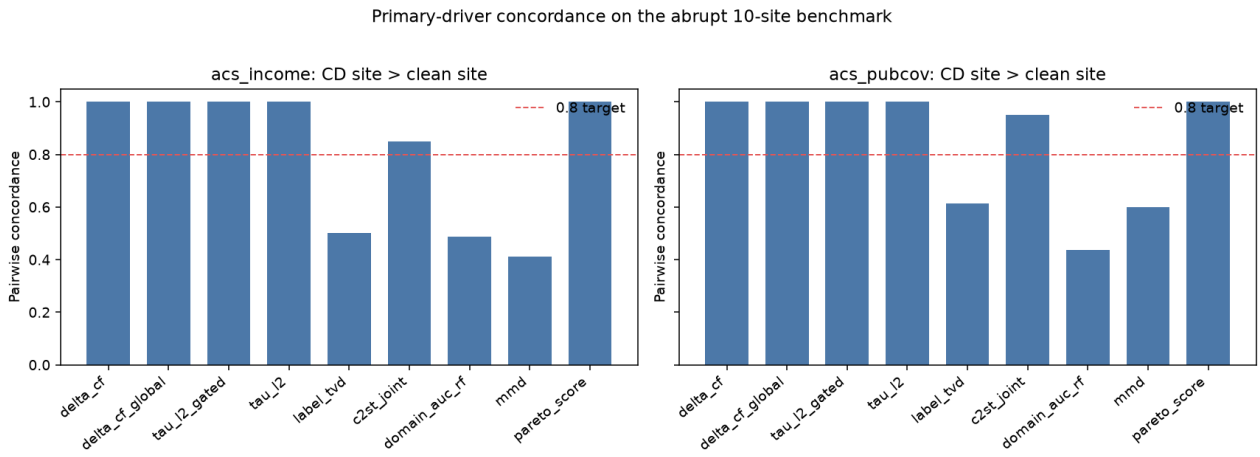


Figure 1: Pairwise concordance against clean sites. The dashed line is the 0.8 target. Δ^{cf}, the whole-window score, skill-gated τ^2, raw τ^2, and the Pareto rank clear the line on both tasks. The joint classifier two-sample test also clears it on average; its failure is the noise contrast in Table 4. Label total variation, the random-forest domain AUC, and RBF MMD do not clear the line. Source: abrupt activation, $\alpha = 1$, $q = 0.5$, four seeds.

them on Income and five eighths on Public Coverage. No clean site and no noise site is undominated. Linear sites are often dominated by a nonlinear site, because the nonlinear rewrite covers every row and the linear rewrite covers a tail, but they remain strictly ahead of every clean site. That is the sense in which the Pareto order puts primary sources first. It does not claim that every primary source lies on the front.

1.4 Time localization

The curve at site k reuses the burn-in reference and recomputes (4) on a sliding window of four batches, using only the whole window and the upper tail. The horizontal coordinate in Figure 2 is the last batch included in the window. Drift begins at batch 8. A window of length four is entirely inside the drift only when its last batch is at least 11. The figure follows that arithmetic. Means for clean sites and for the noise site stay below 0.01 on both sides of $t^\star$. Linear and nonlinear means leave zero between batches 9 and 10 and level off at batch 11. On Income the nonlinear plateau is near 0.32 and the linear plateau is near 0.22. On Public Coverage the two plateaus sit together near 0.18, which is consistent with the similar headline scores in Table 3.

Two detectors are applied to each curve without being told $t^\star$. The online rule calibrates on the first two evaluated endpoints, sets the threshold at that maximum plus 0.01, and reports the first later endpoint that exceeds the threshold. A hit is an estimate within two batches of $t^\star$. The derivative rule reports the endpoint of the largest first difference. Table 5 shows that the derivative is estimating the onset of the plateau rather than the onset of the drift. Its mean absolute error is about three batches on every structured type, which is one window length. The online rule is the onset estimator. It hits every nonlinear site on both tasks and every linear site on Public Coverage. Income linear sites, whose tail is the thin one, hit in 0.833 of replicates, with mean absolute error 1.92 batches. The cost is a false alarm rate of 0.375 on Income clean sites and 0.125 on Public Coverage clean sites: any crossing counts, whether or not it falls near $t^\star$. Among the Income clean curves that do cross, the mean absolute error is 4.17 batches, so the typical alarm is not a call at the change. A threshold of 0.01 above a two-point calibration is enough to rank sites when the window is the

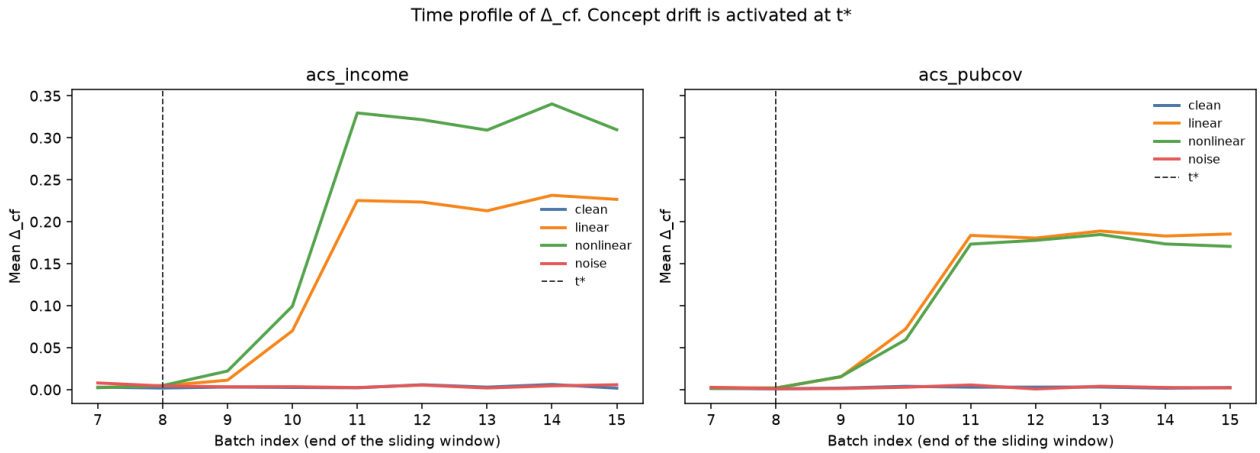


Figure 2: Mean Δ^{cf} against the last batch in a window of length four. The vertical line is $t^\star = 8$. The curve levels off at batch 11, the first endpoint whose window lies entirely after the change. Clean and noise traces remain near zero. Source: abrupt activation, $\alpha = 1$, $q = 0.5$, four seeds.

Table 5: Time localization over four seeds. Online hit rate is the fraction of sites whose first threshold crossing lies within two batches of $t^\star$. False-alarm rate is computed only for clean sites. Derivative MAE is the absolute error of the largest first difference.

Task	Type	Online hit	False alarm	Online MAE	Derivative MAE
Income	Clean	0.062	0.375	4.17	4.06
Income	Linear	0.833	—	1.92	2.83
Income	Nonlinear	1.000	—	1.62	3.00
Income	Noise	0.250	—	4.00	3.25
Public Coverage	Clean	0.062	0.125	3.00	3.88
Public Coverage	Linear	1.000	—	1.42	3.00
Public Coverage	Nonlinear	1.000	—	1.50	3.00
Public Coverage	Noise	0.000	—	3.00	4.00

full recent block, and it is not enough to call a changepoint on a short window without a second screen. The noise site crosses the threshold in two of four Income replicates and in one of four Public Coverage replicates. One Income crossing lands on batch 10, inside the two-batch window; the other is six batches later. The Public Coverage crossing is three batches after $t^\star$ and is not counted as a hit.

1.5 Concept-group type resolution

Detecting a primary source does not say whether the new concept is the linear halfspace or one of the two nonlinear rules. The dictionary is fixed before looking at site labels. On a window with outcome y, three logistic regressions are cross-fit against an intercept-only Brier score: (Z_1, Z_2), the scalar $Z_1 Z_2$, and $(\sin Z_1,\ Z_3^2)$. Group importance is the drop in Brier. The call is the maximizing group. One call uses every row in the recent window. The other uses only the upper tail of X_1, the region in which the linear rule was planted. Accuracy is computed on the five structured sites only. Top-five precision asks a different question: of the five sites with the largest Δ^{cf}, how many lie in $\mathcal{P}$.

Table 6 separates detection from typing. Top-five precision is 1 in every seed: the five largest

Table 6: Type resolution at $\alpha = 1$, abrupt activation, and $q = 0.5$. Accuracy is the fraction of structured sites whose maximizing concept group equals the planted mechanism, averaged over four seeds. Recall is pooled over both tasks and all four seeds, hence over 24 linear sites and 8 sites of each nonlinear mechanism. Top-five precision of Δ^{cf} is 1 in every seed on both tasks.

Call	Mechanism or task	Accuracy or recall
All rows	Income, accuracy	0.700
All rows	Public Coverage, accuracy	0.700
Upper tail	Income, accuracy	0.750
Upper tail	Public Coverage, accuracy	0.900
All rows	Linear, recall	0.500
All rows	Product, recall	1.000
All rows	Sine-square, recall	1.000
Upper tail	Linear, recall	0.917
Upper tail	Product, recall	0.375
Upper tail	Sine-square, recall	1.000

values of Δ^{cf} are exactly the five structured sites. Conditional on that detection, the full-window dictionary recovers both nonlinear mechanisms in every replicate and recovers the linear halfspace in half of the linear sites. The miss is structural. Outside the tail the linear site still carries y^{base}, and on Income that baseline outcome is already monotone in hours and schooling. An additive logistic fit can look competitive with the product and sine-square groups even when the rewritten slice is a halfspace in (Z_1, Z_2). Restricting the call to the upper tail raises linear recall from 0.500 to 0.917 and raises Public Coverage accuracy from 0.700 to 0.900. The same restriction lowers product recall from 1 to 0.375. The product rule was planted on every row. Inside the upper tail the factors are truncated, and the additive group in (Z_1, Z_2) is no longer dominated by the product scalar. The sine-square group remains separated on either window. Type resolution is therefore a property of the region on which the group is scored, not a single accuracy number.

1.6 Feature support and one-shot aggregation

Feature scores are computed on the recent window of each structured site and compared with that site's planted support by AUC. Four scores are used. Shift permutation importance is the drop in the squared concept contrast when a coordinate is permuted inside five quantile bins of the covariate it is most correlated with. Both the local forest and the reference forest are re-evaluated on that same permuted matrix. The local forest here is fit on the full recent window; the cross-fit in (4) is not reused. Predictive permutation importance is the drop in held-out AUC for predicting the recent label. The difference of that importance between the recent window and the burn-in is the fourth score's companion; the fourth score itself is the absolute coefficient vector of an ℓ_1-penalized logistic regression on the recent window, with $C = 0.5$.

Table 7 says that the planted coordinates are recoverable from a single drifted site. Predictive permutation AUC is at least 0.927. Shift permutation AUC is at least 0.906 on linear sites and falls to 0.789 on Income nonlinear sites, where the product and the sine-square rule are functions of coordinates that also predict the baseline income label. The ℓ_1 coefficient is weaker than the forest on linear Public Coverage (0.841) and stronger than shift permutation on Income nonlinear sites (0.930). None of these AUCs is a site-ranking result. They are the contents of the vector a site would upload.

Table 7: Mean AUC of a feature score against the planted support. Linear and product supports are $\{X_1, X_2\}$. The sine-square support is $\{X_1, X_3\}$. Averages run over the structured sites of four seeds.

Task	Type	Shift permutation	Predictive permutation	ℓ_1 coefficient
Income	Linear	0.906	0.927	0.786
Income	Nonlinear	0.789	1.000	0.930
Public Coverage	Linear	0.953	0.995	0.841
Public Coverage	Nonlinear	0.985	1.000	0.879

Aggregation treats each upload as a point on the simplex. Negative shift-permutation coordinates are truncated at zero and the vector is renormalized. The target vector puts mass on a coordinate proportional to the number of structured sites that plant it. Uniform averaging, the coordinate-wise median, inverse-variance weights with variance estimated from the permutation repeats and floored at 10^{-4}, and weights proportional to Δ^{cf} are compared by AUC, precision at the number of positive target coordinates, and Spearman correlation with the target. Leaving the noise site out of the uniform average, and averaging only the five structured sites, are oracles used as ceilings rather than as methods.

On Public Coverage every shift aggregator has AUC at least 0.984, and the Δ^{cf}-weighted average and the average that drops the noise site both have AUC 1 and precision 1. On Income the uniform shift average has AUC 0.869 and precision 0.667. Weighting by Δ^{cf} raises those figures to 0.917 and 0.750. The predictive permutation upload and the ℓ_1 upload, averaged uniformly, reach AUC 0.988 on Income. That is not evidence against the concept score. On Income the planted coordinates are hours, schooling, and age, which are also the coordinates that predict y^{base} on a clean site. A clean site's predictive vector already points at the target. The shift vector is the one that is trying to upload the change rather than the baseline regression, and it is the vector that improves when clean and noise sites are down-weighted by Δ^{cf}.

Leave-one-site-out influence, defined as one minus the Spearman correlation between the uniform aggregate and the aggregate with one site removed, does not identify primary sources. Mean influence on Income is 0.033 for clean sites, 0.013 for linear sites, 0.018 for the noise site, and 0.021 for nonlinear sites. On Public Coverage every type mean lies between 0.005 and 0.008. The uploaded vectors agree with each other closely enough that deleting one site barely moves the ranking. Site-level Δ^{cf} is the functional that separates primary sources from clean sites. Influence on a simplex average is not.

1.7 Intensity, tail mass, and gradual activation

The main grid holds $\alpha = 1$, abrupt activation, and $q = 0.5$. The sensitivity grid varies those three levers on Income only, with three seeds and the same score, and it does not recompute feature uploads or the joint two-sample test. Because the 0.5 and 0.7 quantiles of hours worked are the same mass point at 40 hours, $q = 0.7$ selects the same tail as $q = 0.5$ and is omitted. The quantile 0.8 moves the threshold to 45 hours and leaves about 17% of rows in the linear tail. The quantile 0.9 moves it to 50 hours and leaves about 8%. Nonlinear sites continue to rewrite every activated row. Only the linear sites become thinner.

Table 8 and Figure 3 locate the boundary of the 0.8 target. At full intensity the concordance remains 1 down to a linear tail of about 17% of rows, whether the change is abrupt or spread over four batches. At a tail of about 8%, full abrupt intensity falls to mean concordance 0.850 with a worst seed of 0.750, so the mean still clears 0.8 and the guarantee across seeds does not. Gradual activation at that thin tail has the same character (0.867, worst seed 0.800). Lowering the rewrite

Table 8: Income concordance $C(\Delta^{\mathrm{cf}}; \mathcal{P}, \mathcal{C})$ under weaker rewrites and thinner linear tails. Three seeds. The worst seed is in parentheses. Quantile 0.5 is the main tail, about 23% of rows because of the mass at 40 hours.

	$q = 0.5$		$q = 0.8$		$q = 0.9$	
α	Abrupt	Gradual	Abrupt	Gradual	Abrupt	Gradual
1.0	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)	1.000 (1.000)	0.850 (0.750)	0.867 (0.800)
0.7	1.000 (1.000)	1.000 (1.000)	0.967 (0.950)	0.950 (0.900)	0.817 (0.750)	0.683 (0.500)
0.4	0.783 (0.650)	0.750 (0.650)	0.617 (0.450)	0.600 (0.450)	0.550 (0.450)	0.550 (0.450)

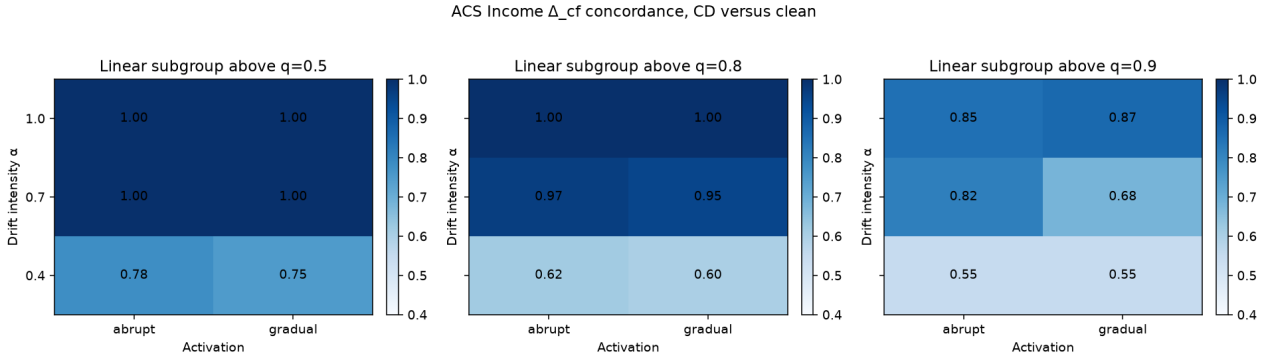


Figure 3: Income concordance of Δ^{cf} against clean sites. Each panel is one linear-tail quantile. Cell entries are means over three seeds. Means stay at or above 0.8 for $\alpha \in \{0.7, 1\}$ down to a linear tail of about 17% of rows. They fall below 0.8 when $\alpha = 0.4$, and when a tail of about 8% is combined with $\alpha = 0.7$ and a gradual ramp. Full intensity on that thin tail stays above 0.8 on average and not in every seed; the worst seeds are in Table 8.

probability to 0.7 preserves the target on the main tail and on the 17% tail. Combining $\alpha = 0.7$ with the 8% tail and a four-batch ramp drops the concordance to 0.683, with a worst seed of 0.500. At $\alpha = 0.4$ every column is below 0.8. The main-tail abrupt entry is 0.783 with a worst seed of 0.650. On that same $\alpha = 0.4$ slice, the whole-window score without the tail scan has concordance 0.433. The region maximum is what keeps a weak but structured rewrite near the target. It does not keep an arbitrarily weak rewrite above the target.

1.8 What the benchmark supports

The benchmark supports four claims, each tied to a table. First, when the rewrite is complete and the linear tail contains about a fifth of the rows or more, Δ^{cf} orders every structured site ahead of every clean site and ahead of the noise site, on two real covariate tables, in every seed (Table 4). Second, the functionals that ignore either the existence of a local concept or the region in which that concept lives do not. Label total variation and the covariate arms are near chance against clean sites. The joint two-sample test and the pooled Brier excess promote the noise site, completely so for linear sites under the pooled excess on Income. Third, the time profile of the same score locates an abrupt change to within about two batches once the scoring window is taken into account, with a false-alarm rate on clean Income sites that a deployment threshold would still have to budget for (Table 5). Fourth, a pre-specified concept-group dictionary can name the nonlinear mechanisms from the full window and the linear mechanism from the tail, and it cannot do both from a single region

(Table 6).

The benchmark does not support a claim that leave-one-site-out influence on an averaged variable-importance vector identifies the primary site. Those vectors are too similar across sites for the deletion of one upload to move the rank correlation. It also does not support a claim at very small intensity. Below a rewrite probability of one half, or inside a linear tail that covers less than a tenth of the window, pairwise concordance falls through 0.8 on at least one seed. Both limits are visible in the same design that produces the perfect separation at full intensity, which is the reason for reporting them beside the headline rather than as a separate simulation.